

Lecture 4, Jan 28-2026.

* Derivative of the Natural Logarithm function.

Thm: $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$

Pf. Let $y = \ln(x)$ then we know

$$e^y = x$$

by implicit derivative we have:

$$\frac{d}{dx}(e^y) = \frac{d}{dx}(x)$$

$$e^y \cdot \frac{dy}{dx} = 1 \quad \text{then,}$$

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

Example. Differentiate $y = \ln(x^3+1)$.

Sln. By the chain rule.

$$y' = \frac{1}{x^3+1} \cdot \frac{d}{dx}(x^3+1) = \frac{3x^2}{x^3+1}.$$

Remk. $\frac{d}{dx}[\ln(g(x))] = \frac{g'(x)}{g(x)}$

Example. Find $\frac{d}{dx}(\ln(\sin x))$.

Sln. $\frac{d}{dx}(\ln(\sin x)) = \frac{\cos x}{\sin x} = \cot x.$

Example. Differentiate $f(x) = \sqrt{\ln x}$.

Sln. $f'(x) = \frac{1}{2x\sqrt{\ln x}}.$

Example. Find $\frac{d}{dx} \left(\ln \left(\frac{x+1}{\sqrt{x-2}} \right) \right)$.

Sln.1

$$\begin{aligned} \frac{d}{dx} \left(\ln \left(\frac{x+1}{\sqrt{x-2}} \right) \right) &= \frac{1}{\left(\frac{x+1}{\sqrt{x-2}} \right)} \cdot \frac{d}{dx} \left(\frac{x+1}{\sqrt{x-2}} \right) \\ &= \frac{\sqrt{x-2}}{x+1} \cdot \frac{d}{dx} \left(\frac{x+1}{\sqrt{x-2}} \right) \\ \frac{d}{dx} \left(\frac{x+1}{\sqrt{x-2}} \right) &= \frac{1 \cdot (\sqrt{x-2}) - (x+1) \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{x-2}}}{(\sqrt{x-2})^2} \\ &= \frac{2(x-2) - (x+1)}{(x-2)^2 \sqrt{x-2}} \\ &= \frac{2x-4-x-1}{2(x-2)\sqrt{x-2}} = \frac{x-5}{2(x-2)\sqrt{x-2}}. \end{aligned}$$

So,

$$\frac{d}{dx} \left(\ln \left(\frac{x+1}{\sqrt{x-2}} \right) \right) = \frac{\cancel{\sqrt{x-2}}}{x+1} \cdot \frac{x-5}{2(x-2)\cancel{\sqrt{x-2}}} = \frac{x-5}{2(x-2)(x+1)}.$$

Sln.2

$$\begin{aligned} \ln \left(\frac{x+1}{\sqrt{x-2}} \right) &= \ln(x+1) - \ln(\sqrt{x-2}) \text{ by } \ln\left(\frac{x}{y}\right) = \ln x - \ln y. \\ &= \ln(x+1) - \frac{1}{2} \ln(x-2). \text{ by } \ln(x^m) = m \cdot \ln(x). \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} \ln \left(\frac{x+1}{\sqrt{x-2}} \right) &= \frac{d}{dx} \left[\ln(x+1) - \frac{1}{2} \ln(x-2) \right] \\ &= \frac{d}{dx} \ln(x+1) - \frac{1}{2} \frac{d}{dx} \ln(x-2) \\ &= \frac{1}{x+1} - \frac{1}{2} \cdot \frac{1}{x-2} \\ &= \frac{2(x-2) - (x+1)}{2(x-2)(x+1)} = \frac{2x-4-x-1}{2(x-2)(x+1)} = \frac{x-5}{2(x-2)(x+1)} \end{aligned}$$

Example. Find the absolute minima of $f(x) = x^2 \ln x$.

Sln. Dom $f = (0, \infty)$. By the product rule

$$f'(x) = 2x \ln(x) + x^2 \cdot \frac{1}{x} = 2x \ln x + x = x(2 \ln x + 1)$$

$$f'(x) = 0 \Leftrightarrow x = 0 \text{ or } 2 \ln x + 1 = 0$$

$$\ln x = -\frac{1}{2}$$

$$x = e^{-1/2}$$

So $f'(x) > 0$ then f is increasing so, $f(e^{-1/2})$ is max.

Example. Find $f'(x)$ if $f(x) = \ln|x|$

Sln. Since

$$f(x) = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$

it follows that

$$\begin{aligned} f'(x) &= \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{-x} \cdot (-1) & \text{if } x < 0 \end{cases} \\ &= \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{x} & \text{if } x < 0 \end{cases} \end{aligned}$$

Thm.

$$\frac{d}{dx} \ln|x| = \frac{1}{x}.$$

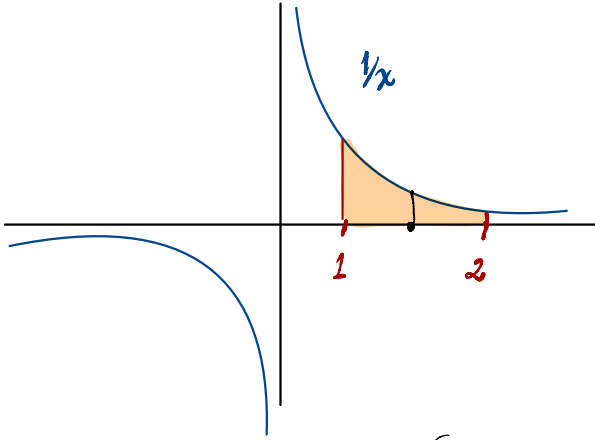
Thm.

$$\int \frac{1}{x} dx = \ln|x| + C.$$

Example. Find the area under the hyperbola $xy=1$ from $x=1$ to $x=2$.

Soln. Notice that $xy=1 \Leftrightarrow y=\frac{1}{x}$. So,

$$A = \int_1^2 \frac{1}{x} dx = \ln x \Big|_1^2 = \ln(2) - \ln(1) = \ln(2) \approx 0.693.$$



Example. Evaluate $\int \frac{x}{x^2+1} dx$.

Sln. $\int x \cdot \frac{1}{x^2+1} dx.$ $f(x) = \frac{1}{x^2+1} = h(g(x))$ $h(x) = \frac{1}{x}$
 $g(x) = x^2+1$

$g'(x) = 2x$. let $u = x^2 + 1$ $du = 2x dx \Leftrightarrow \frac{1}{2} du = x dx$.
by substitution. $dx = \frac{1}{2x} du$

$$\int x \cdot \frac{1}{x^2+1} dx = \int \frac{1}{2u} du = \int \cancel{x} \cdot \frac{1}{u} \cdot \frac{du}{\cancel{2x}} = \int \frac{du}{2u}$$

$$= \frac{1}{2} \int \frac{1}{u} \cdot du$$

$$= \frac{1}{2} \ln |u| + C.$$

$$= \frac{1}{2} \ln |x^2 + 1| + C.$$

$$= \frac{1}{2} \ln(x^2+1) + C \quad \ln(x^m) = m \ln(x)$$

$$= \ln(\sqrt{x^2+1}) + C.$$

Example. Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Sln. $\int \frac{\ln x}{x} dx = \int \frac{1}{x} \cdot \ln x dx$

$f(x) = \ln x$
 $f(x) = h(g(x))$ $h(x) = x$
 $g(x) = \ln x$
 $g'(x) = \frac{1}{x}$

let $u = \ln x$ $du = \frac{1}{x} \cdot dx$

$$\int \frac{\ln x}{x} dx = \int u \cdot du = \frac{u^2}{2} + C = \frac{1}{2} \ln^2(x) + C$$

$$\begin{aligned} \int_1^e \frac{\ln x}{x} dx &= \left. \frac{1}{2} \ln^2(x) \right|_1^e = \frac{1}{2} (\ln^2(e) - \ln^2(1)) \\ &= \frac{1}{2} (1 - 0) = \frac{1}{2} \end{aligned}$$

Sln. 2.

$$\begin{aligned} \int_1^e \frac{\ln x}{x} dx & \quad u = \ln(x) \quad x=1 \Rightarrow u=0 \\ & \quad x=e \Rightarrow u=1 \\ &= \int_0^1 u du = \left. \frac{u^2}{2} \right|_0^1 = \frac{1}{2} - 0 = \frac{1}{2} \end{aligned}$$

Example. Evaluate $\int \tan x dx$.

* General Logarithmic and Exponential functions.

$$\log_b(x) = \frac{\ln(x)}{\ln(b)}, \quad b > 1.$$

$$\frac{d}{dx} (\log_b(x)) = \frac{1}{\ln b} \cdot \frac{d}{dx} (\ln x) = \frac{1}{\ln(b)x}$$

$$\frac{d}{dx} (b^x) = b^x \cdot \ln(b)$$

Also we have:

$$\int b^x dx = \frac{b^x}{\ln(b)} + C.$$

Goal: Compute $\frac{d}{dx}(x^{\sqrt{x}})$.